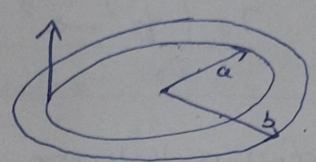
# Numericals

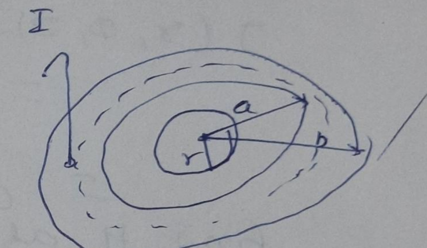
Q. Using Ampere's circuital law in Integral form, find H everywhere due to the following volume current density in cylindrical coordinates.

$$J = \begin{cases} = 0 & 0 < r < a \\ = J_0 I_z & a < r < b \\ = 0 & b < r < \infty \end{cases}$$




Sol/n

Using Ampere law




$\oint H \cdot dl = I_{enclosed}$

for $r \leq a$ (Ampere loop)

No current $I = 0$

$H = 0 \quad (r < a)$

for $b < r > a$

$\int H \cdot dl = I_{enc}$

$\int H \cdot (\cancel{dr} + r d\phi + \cancel{dz}) = \int J \cdot ds$

$$\int H \cdot 2\pi r = \int_{\phi=0}^{2\pi} \int_{r=a}^{r} J_0 \, r \, d\phi \, dr$$

$$H (2 \cancel{\pi} r) = \frac{J_0 \cancel{2\pi} (r^2 - a^2)}{2}$$

$$H = \frac{J_0 (r^2 - a^2)}{2r} I_\phi \quad \text{(for } r > b)$$

for $r \geq b$

$$H = \frac{J_0 (b^2 - a^2)}{2b} I_\phi$$

$$\oint H \cdot dl = \int J \cdot ds$$

$$\cancel{H(2\pi r) = \iint r\,dr\,d\phi}$$

$$\cancel{H(2\pi r) = 2\pi}$$

$$H(2\pi r) = J_0 \pi (b^2 - a^2)$$

$$\boxed{H = \frac{J_0 (b^2 - a^2)}{2r} I_\phi}$$

Q Consider a volume current density distribution in cylindrical coordinates.

$$J(r, \phi, z) = 0 \qquad 0 < r < a$$

$$= J_0 \left(\frac{r}{a}\right) I_z \qquad a < r < b$$

$$= 0 \qquad b < r < \infty$$

find H at various region

Soln for $r < a$

$H = 0$

for $a < r < b$

$$H \cdot 2\pi r = \int_{\phi=0}^{2\pi} \int_{r=a}^{r} (J_0) \left(\frac{r}{a}\right) r\,dr\,d\phi$$

$$H \cdot 2\cancel{\pi} r = \frac{J_0}{3a} \left(r^3\right)_a^r 2\pi$$

$$H \cdot (\cancel{2\pi} r) = \frac{J_0}{3a} (r^3 - a^3)$$

$$H = \frac{J_0}{3ar} (r^3 - a^3) I_\phi$$

at $r = b$ ; $\boxed{\frac{J_0}{3ab} (b^3 - a^3) I_\phi = H}$

Region $b < r < \infty$ , $J = 0$

$\oint H \cdot dl = \cancel{I_\phi}\ H(2\pi r) = \frac{J_0}{3ab}(b^3 - a^3)\ I_\phi$

$H(2\pi r) = \frac{2\pi J_0}{3a}(b^3 - a^3)$

$H = \frac{J_0}{3ar}(b^3 - a^3)\ I_\phi$

for region $a < r < b$ find vector Potential A

in a region

$H = \frac{J_0}{3ar}(r^3 - a^3)\ I_\phi$

$I_\phi$ $B = \nabla \times A$

$$\nabla \times A = \begin{vmatrix} \frac{I_r}{r} & I_\phi & \frac{I_z}{z} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_r & r A_\phi & A_z \end{vmatrix}$$

As wire in z direction $(A_r = 0)$, $A_\phi = 0$

$-I_\phi \left| \frac{\partial}{\partial r} A_z - \cancel{\frac{\partial}{\partial z} A_r} \right|$

$\nabla \times A = -I_\phi \left( \frac{\partial}{\partial r} A_z \right) = B$

$\frac{d A_z}{\cancel{dr}} = \int_{r=a}^{r} \frac{-\mu_0 J_0}{3ar}(r^3 - a^3)\, dr$

$A_z = -\frac{\mu_0 J_0}{3a}\left[ \left( \frac{r^3}{3} \right)_a^r - a^3 \ln r \Big|_a^r \right]$

$$A_z = \frac{\mu_0 J_0}{3a}\left[\left(a^3 \ln \frac{r}{a} - \left(\frac{r^3 - a^3}{3}\right)\right)\right] I_z$$

Q2 Given magnetic vector potential $A = 10 \sin\theta \, I_\phi$ in spherical coordinates. find the flux density at $(2, \pi/2, 0)$

(b) Given vector magnetic potential $A = 5e^{-r}\cos\phi \, I_r - 5\cos\phi \, I_z$ in cylindrical coordinate. Find flux density at $\left(2, \frac{3\pi}{2}, 0\right)$

Sol/n

$$B = \nabla \times A$$

$$B = \nabla \times A = \begin{vmatrix} \frac{I_r}{r^2 \sin\theta} & \frac{I_\theta}{r \sin\theta} & \frac{I_\phi}{r} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ 0 & 10 r \sin\theta & 0 \end{vmatrix}$$

$$= \frac{\partial}{\partial r}(10 r \sin\theta) \cdot \frac{I_\phi}{r}$$

$$B = \frac{10 \sin\theta}{r} I_\phi$$

$$B\left(2, \frac{\pi}{2}, 0\right) = \frac{10}{2} I_\phi = 5 I_\phi$$

$$B = \nabla \times A$$

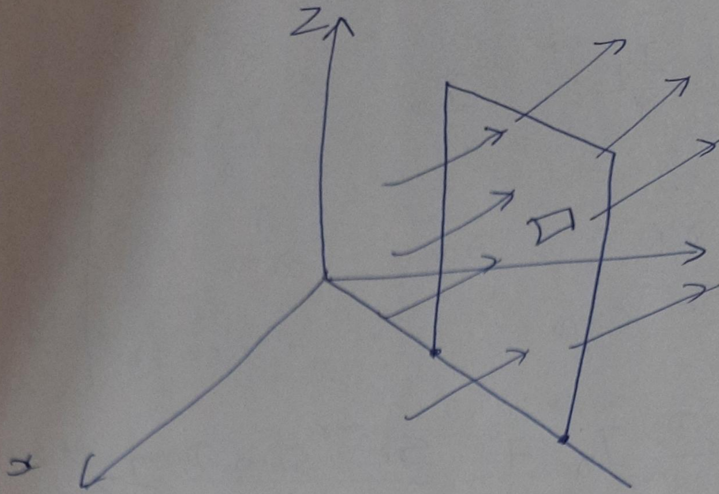
$$B = \nabla \times A = \begin{vmatrix} \frac{I_r}{r} & I_\phi & \frac{I_z}{r} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ 5e^{-r}\cos\phi & 0 & -5\cos\phi \end{vmatrix}$$

$$B = \frac{5\sin\phi}{r} I_r + \frac{5e^{-r}\sin\phi}{r} I_z$$

$$B\left(2, \frac{3\pi}{2}, 0\right) = \frac{5}{2}\sin\frac{3\pi}{2} I_r + \frac{5e^{-2}}{2}\sin\frac{3\pi}{2} I_z$$

$$\boxed{B = -2.5\, I_r - 0.338\, I_z}$$

Q. The flux density at a point distance $r$ from a long filamentary conductor carrying a current $I$ in $I_z$ direction is given by $B = \frac{\mu_0 I}{2\pi r} I_\phi$. Calculate the flux crossing the portion of the plane $\phi = \frac{\pi}{4}$ defined by $0.04 < r < 0.05$ m and $0 < z < 2$ m for current 2A




S

Soln

$$\Phi_m = \int B \cdot ds$$

$$\Phi_m = \int_{z=0}^{2} \int_{r=0.01}^{r=0.05} \left(\frac{\mu_0 I}{2\pi r}\right)(dr\,dz)\, I_\phi$$

$$\Phi_m = \frac{\cancel{2}\mu_0 I}{\cancel{2}\pi} \ln r \Big|_{0.01}^{0.05}$$

$$\Phi_m = \frac{4\pi \times 10^{-7}}{\pi} \times 2 \times \ln \frac{0.05}{0.01}$$

$$\boxed{\Phi_m = 1.29\,\mu\,Wb}$$

Q4 There exist a boundary b/w two magnetic material at $z=0$, having permeability $\mu_1 = 4\mu_0$ H/m for region 1 where $z>0$ and $\mu_2 = 7\mu_0$ H/m for region 2 where $z<0$. There exist a surface current of density $K = 60\, I_x$ A/m at the boundary $z=0$. For a field

$$B_1 = 2I_x - 3I_y + 2I_z \text{ mT in region 1}$$

find value of flux density ($B_2$) in region 2

Sol/n

$$B_1 = (2I_x - 3I_y + 2I_z) \times 10^{-3}$$

$$B_{n1} = B_{n2} = 2I_z \times 10^{-3} T$$

$$B_1 = B_{n1} + B_{t1} \quad , \quad B_2 = B_{n2} + B_{t2}$$

$$B_{t1} = 2I_x - 3I_y$$

$$H_{t1} = \frac{(2I_x - 3I_y) \times 10^{-3}}{4 \times 4\pi \times 10^{-7}}$$

$$B_{n1} = B_{n2}$$

$$H_{t1} - H_{t2} = K I_n$$

$$H_{t1} = 398\, I_x - 597\, I_y \text{ A/m}$$

x

z y

$$H_{t1} - K \times I_n = H_{t2}$$

$$(398 I_x - 597 I_y) - 60 I_x \times I_z$$

$$= 398 I_x - 597 I_y - 60(-I_y)$$

$$= 398 I_x - 537 I_y$$

$$B_{t2} = \mu_2 H_{t2} = 3.5 I_x - 4.72 I_y \text{ mT}$$

$$B_2 = 3.5 I_x - 4.72 I_y + 2 I_z$$

Q $H = 20 \frac{(x I_x + y I_y)}{x^2 + y^2}$ A/m

Show $\nabla \cdot B = 0$

find current density J

(C) find scalar vector potential $V_m(x, y, z)$ if $V_m = 0$ at (1, 1, 1)

Soln

$$\nabla \cdot H = 0 \Rightarrow \frac{\partial}{\partial x}\left(\frac{20x}{x^2+y^2}\right) + \frac{\partial}{\partial y}\left(\frac{20y}{x^2+y^2}\right)$$

$$\frac{\partial}{\partial x} \cdot \frac{x}{x^2+y^2} + \frac{\partial}{\partial y} \cdot \frac{y}{x^2+y^2}$$

$$\frac{(x^2+y^2)\cdot 1 - x \cdot 2x}{(x^2+y^2)^2} + \frac{(x^2+y^2)\cdot 1 - y(2y)}{(x^2+y^2)^2}$$

$$\cancel{\frac{y^2 - x^2}{(x^2+y^2)^2}} + \cancel{\frac{x^2 - y^2}{(x^2+y^2)^2}} = 0$$

$$\int H \cdot dl = \int J \cdot ds = 0$$

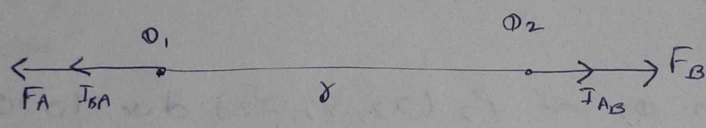
Electrostatics Unit-2

$$F \propto \frac{Q_1 Q_2}{r^2}$$

$$F = k \frac{Q_1 Q_2}{r^2}$$

$$k = \frac{1}{4\pi\varepsilon} \quad ; \quad \varepsilon_0 = 8.854 \times 10^{-12}\ F/m$$


Coulamb law in vector form

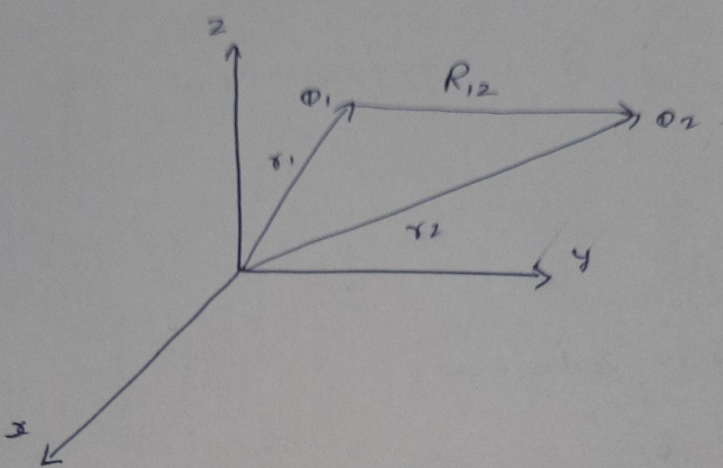



Coulamb law is linear. The force at a Point charge due to two or more charges can be ~~place~~ added vectorially

$$F = \frac{1}{4\pi\varepsilon_0} \frac{Q_1 Q_2}{r^2} I_r$$

$I_r$ is a unit vector in $r$ direction represent a force of charge $Q_2$ by charge $Q_1$, which is considered away from the charge, Indicating the direction of force

A point charge $Q_2$ is located in vaccum at point $P_2\,(x_2, y_2, z_2)$. Find the force on charge $Q_2$ due to charge $Q_1$ located at $P_1\,(x_1, y_1, z_1)$. The charge are in coulamb and distance are in meters.




charge on $Q_2$ at $P_2\,(x_2, y_2, z_2)$ due to $Q_1$ at $P_1\,(x_1, y_1, z_1)$ will be

$$F_2 = \frac{Q_1\,Q_2}{4\pi\epsilon_0\,|R_{12}|^2}\,I_{R_{12}}$$

$$R_{12} = \vec{r}_2 - \vec{r}_1$$

$$I_{R_{12}} = \frac{R_{12}}{|R_{12}|} = \frac{\vec{r}_2 - \vec{r}_1}{|r_2 - r_1|}$$

$$|r_2 - r_1| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

$$F_2 = \frac{Q_1\,Q_2}{4\pi\epsilon_0\,|R_{12}|^3}\,(\vec{r}_2 - \vec{r}_1)$$

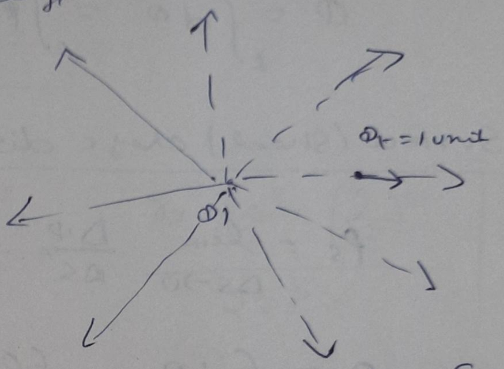
## Electric field Intensity, E

$$F = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_t}{r^2} I_r$$

$Q_2 = Q_t$ ; It is a test charge

$r$ is dist b/w $Q_1$ and $Q_t$

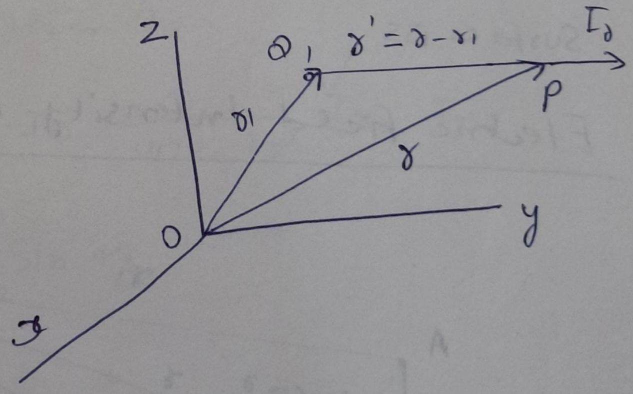
$$\frac{F_t}{Q_t} = E = \frac{Q_1}{4\pi\epsilon_0 r^2} I_r$$




SI Unit of $\vec{E}$ is (N/C)

Force per unit charge is called Electric field Intensity

$$E = \lim_{\Delta Q \to 0} \frac{\Delta F}{\Delta Q}$$




$$E = \frac{Q_1 (r - r_1)}{4\pi\epsilon_0 |r - r_1|^3}$$

## Volume charge distribution

$$\rho_v = \lim_{\Delta v \to 0} \frac{\Delta Q}{\Delta v} \quad C/m^3$$

$$Q = \int_v dQ = \int_v \rho_v \, dv$$

## Line charge distribution

$$\rho_l = \lim_{\Delta l \to 0} \frac{\Delta Q}{\Delta l} \; C/m$$

$$Q = \int_l dQ = \int_l \rho_l \, dl$$

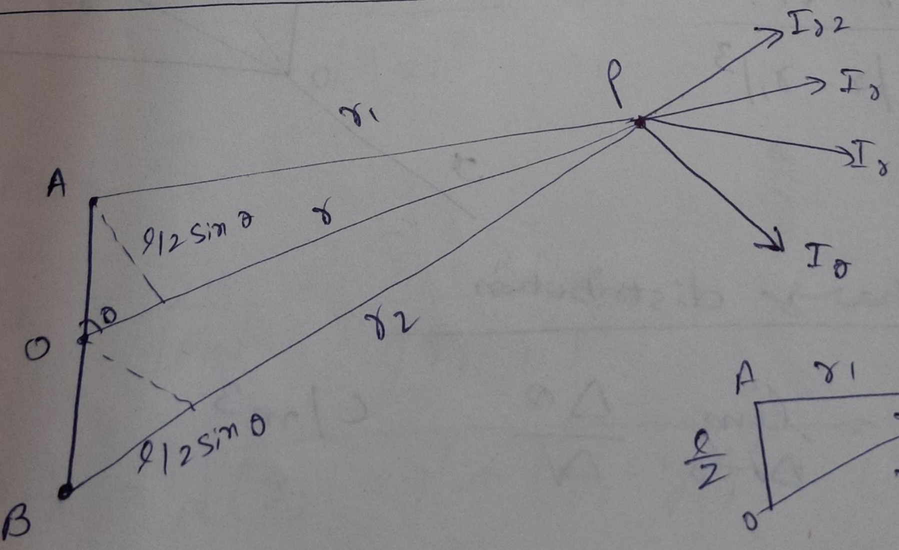
## Surface (sheet) charge distribution

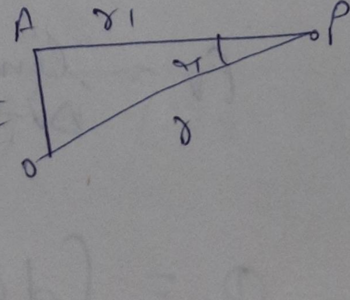
$$\rho_s = \lim_{\Delta S \to 0} \frac{\Delta Q}{\Delta S} \; C/m^2$$

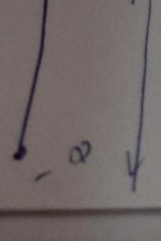
$$Q = \int_S dQ = \int_S \rho_s \cdot ds$$

Where $\Delta Q$ is a small amount of charge on small surface $\Delta S$

## Electric field Intensity due to dipole




$p = Ql$

In $\Delta$ OAP and DOBP

$$OA^2 = \left(\frac{l}{2}\right)^2 = r_1^2 + r_2^2 - 2 r r_1 \cos\alpha_1$$

$$\cos\alpha_1 = \frac{r_1^2 + r^2 - \left(\frac{l}{2}\right)^2}{2 r_1 r}$$

$$\cos\alpha_2 = \frac{r_2^2 + r^2 - \left(\frac{l}{2}\right)^2}{2 r_2 r}$$

$$E_r = \frac{Q}{4\pi\varepsilon_0}\left[\frac{r_1^2 + r^2 - \left(\frac{l}{2}\right)^2}{2 r_1^3 r} - \frac{r_2^2 + r^2 - \left(\frac{l}{2}\right)^2}{2 r_2^3 r}\right]$$

$$E_r = \frac{Q}{2\pi\varepsilon_0 r^3} l\cos\theta \qquad \left[\because r >> l\right) \; r_1 = r - \frac{l}{2}\cos\theta, \quad r_2 = r + \frac{l}{2}\cos\theta$$

The $\theta$ component of E is given by

$$E_\theta = E \cdot I_\theta = \frac{Q}{4\pi\varepsilon_0 r_1^2} I_{r_1} \cdot I_\theta - \frac{Q}{4\pi\varepsilon_0 r_2^2} I_{r_2} \cdot I_\theta$$

$$= \frac{Q}{4\pi\varepsilon_0 r_1^2}\sin\alpha_1 + \frac{Q}{4\pi\varepsilon_0 r_2^2}\sin\alpha_2$$

$$\simeq \frac{Q}{2\pi\varepsilon_0 r^2}\sin\alpha_1$$

$$= \frac{Q}{2\pi\varepsilon_0 r^2}\frac{(l/2)\sin\theta)}{r}$$

$$E_\theta = \frac{Ql}{4\pi\varepsilon_0 r^3}\sin\theta$$

$E = E_r I_\theta + E_\theta I_\theta$

$$E = \frac{Q l}{4\pi\epsilon_0 r^3}(2\cos\theta\, I_r + \sin\theta\, I_\theta)$$

writing $P = Ql$ dipole moment

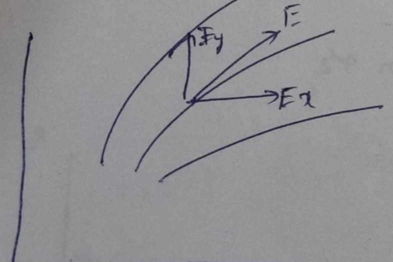
$$E = \frac{P}{4\pi\epsilon_0 r^3}(2\cos\theta\, I_r + \sin\theta\, I_\theta)$$

This is a field due to dipole at a very large distance from dipole.

Dipole moment is in positive z direction [from -ive charge to positive charge.

## Field lines

An electric flux lines is an imaginary path or line drawn in such a way that it's direction at any point is the direction of electric field at that point.




$$\Delta l = \Delta x\, I_x + Dy\, I_y + \Delta z\, I_y$$

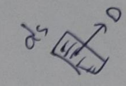
$$\frac{dx}{E_x} = \frac{dy}{E_y} = \frac{dz}{E_z}$$ - Cartesian coordinate

(cylindrical coordinate) $$\frac{dr}{E_r} = \frac{r\, d\Phi}{E_\Phi} = \frac{dz}{I_z}$$ ~~(cylindrical coordinate)~~

(spherical coordinate) $$\frac{dr}{E_r} = \frac{r\, d\theta}{E_\theta} = \frac{r\sin\theta\, d\Phi}{E_\Phi}$$

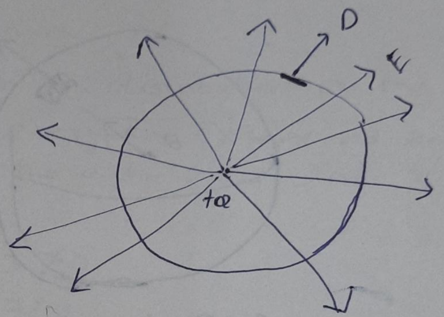
## Electric flux

The electric field at any distance r from a point charge in a free space is expressed a



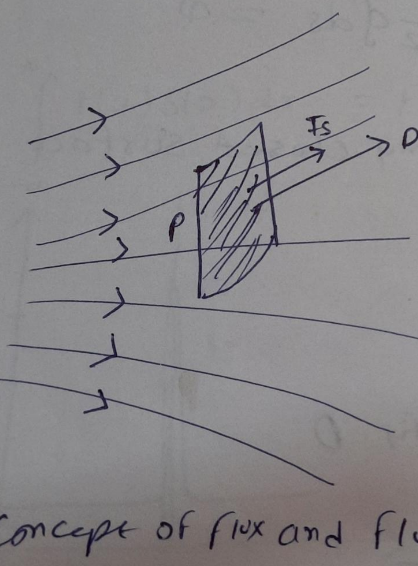
$$E = \frac{Q}{4\pi\varepsilon_0 r^2} \, I_r \; N/C$$

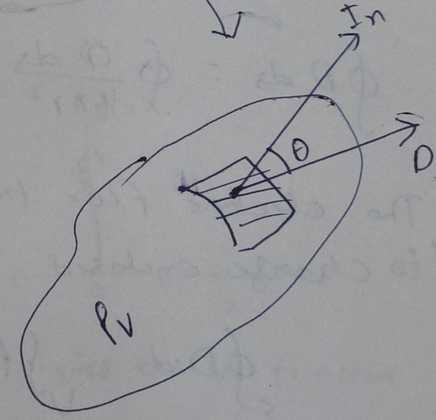
$$\varepsilon_0 E = \frac{Q}{4\pi r^2}$$




$$D = \frac{Q}{4\pi r^2}$$

$$\psi = \int D \cdot ds$$




Concept of flux and flux density

$$d\psi = D\,ds\cos\theta = D \cdot ds \, I_n$$

$$\text{Total flux } \psi = \int_s D \cdot ds$$

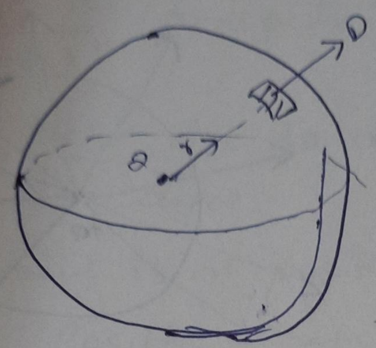
Electric flux density is also called displacement flux density or displacement density

# Gauss law - A Maxwell equation

The total flux out of a closed surface is equal to the net charge within the surface

$$\oint D \cdot ds = Q_{enclosed} = \psi$$

$$E = \frac{Q}{4\pi\epsilon_0 r^2} I_r$$




$$\oint D \cdot ds = \oint \frac{Q \cdot ds}{4\pi r^2} = \frac{Q}{4\pi r^2}\oint ds = Q$$

The electric flux through any closed surface equal to charge enclosed.

$$\oint_S D \cdot ds = \int_V \rho_v \cdot dv$$

Divergence of the flux density D

$$\operatorname{div} D = \lim_{\Delta v \to 0} \frac{\oint D \cdot ds}{\Delta v}$$

$$\lim_{\Delta v \to 0} \frac{\oint D_n \cdot ds}{\Delta v} = \operatorname{div} D = \rho_v \ C/m^3$$

$$\boxed{\operatorname{div} D = \rho_v}$$

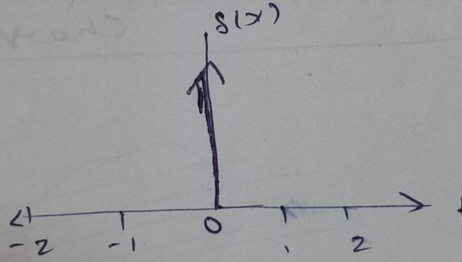
- Gauss law depend on symmetry, if we cannot show that symmetry exist, we cannot use Gauss law
- if Gauss law is not applicable, Coulamb ~~Sto~~ law can be used.

## Dirac delta function

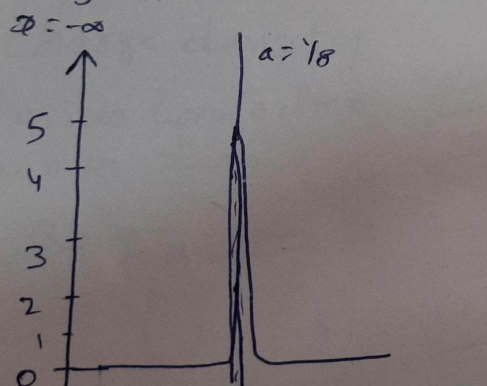
It is a generalized function or distribution on the real no line that is zero everywhere except at zero with an Integral of one over the entire real line.

$$\delta(x) = \begin{cases} 0 & \text{for } x \neq 0 \\ \infty & \text{for } x = 0 \end{cases}$$

$$\int_{x=-\infty}^{\infty} f(x)\,\delta(x)\,dx = f(0)$$



(a) Dirac delta function



(b) dirac-Delta function as the limit of a sequence of zero centred normal distribution

$$\delta_a(x) = \frac{1}{a\sqrt{\pi}} e^{-x^2/a^2}$$

as $a \to 0$

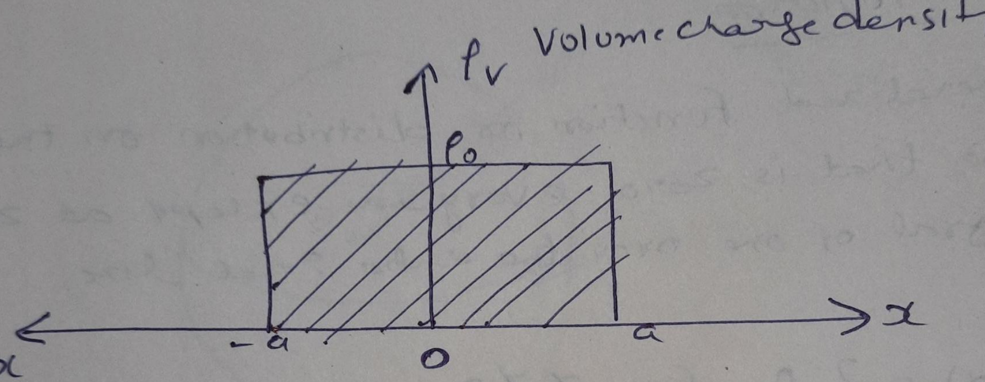
Now In electrostatics, the Volume charge density corresponding to sheet of charge $\rho_{so}$. lying in the $x = 0$ plane is $\rho_{so}\,\delta(x)$

Gauss law in differential form is given by

$$\nabla \cdot E = \frac{1}{\varepsilon_0} \rho_{s0}\, \delta(x)$$

So It can be modified as

$$\nabla \cdot E = \frac{1}{\varepsilon_0} \rho_{s0}\, \delta(x - x_0)$$




(Volume charge density corresponding to Surface charge)

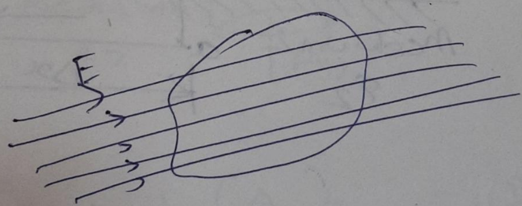
---

Dielectric differ from conductor in that they ~~are~~ have no free charge that can move in the material under the influence of an Electric field. In dielectric all the electrons are bound and the application of electric field causes a minute displacement of positive and negative charge in opposite direction, which creates dipole and material become polarized.

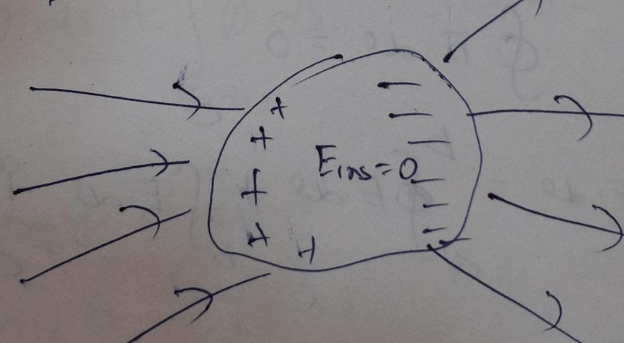
## Conductor in an Uniform Electric field

(1) When a conductor is placed in uniform electric field. The Electric field inside the conductor is zero

(5) Charge density inside conductor will be zero




~~(c) The E surface~~



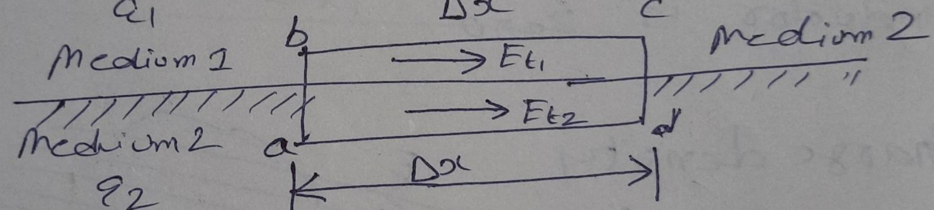

(c) The conductor including the surface, is an equipotential region.

(d) The electric field Intensity at any Point on the surface of the conductor is entirely normal to it and is equal to $\frac{1}{\varepsilon_0}$ times Surface charge density at that Point.

## Boundary Conditions

### In two perfect dielectric media

Field tangent to boundary :




$$\oint E \cdot dl = 0$$

$$\oint E \cdot dl = \int_a^b E \cdot dl + \int_b^c E \cdot dl + \int_c^d E \cdot dl + \int_d^a E \cdot dl = 0$$

at $\Delta y \to 0$

$$\oint E \cdot dl = 0 + E_{t_1} \cdot \Delta x + 0 - E_{t_2} \cdot \Delta x = 0$$

$$E_{t_1} = E_{t_2}$$

The tangential component of E is therefore continous across the boundary

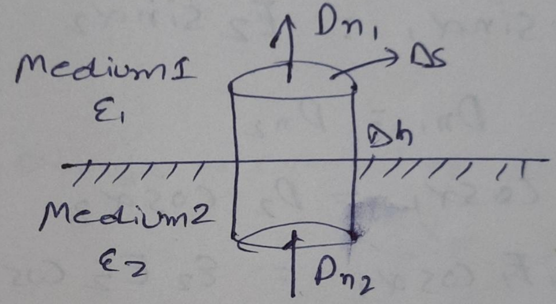
* if boundary lies between dielectric and a conductor then $E = 0$ and in the conductor $E_t = 0$ for both media.

* For static field $E$, is only normal to surface of conductor

## Normal Component of Electric displacement

$$\oint D \cdot ds = Q_{enc}$$

$$\cancel{\oint D \cdot ds + \int D \cdot ds + \int D \cdot ds = Q_{enc}}$$




$$\int_{top} D \cdot ds + \int_{bottom} D \cdot ds + \int_{Sides} D \cdot ds = Q_{enc}$$

$\Delta h \to 0$, $\rho_s$ = Charge density

$$D_{n_1} \Delta s - D_{n_2} \Delta s = \rho_s \Delta s$$

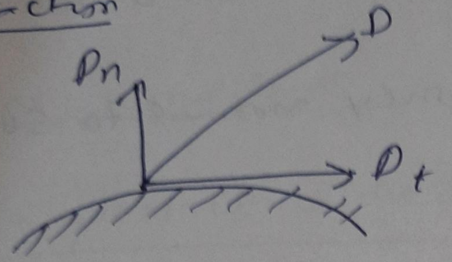
$$\boxed{D_{n_1} - D_{n_2} = \rho_s}$$

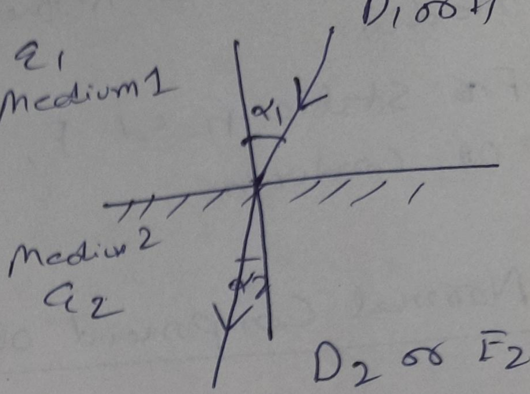
* At a dielectric to dielectric boundary, the difference of normal component of electric displacement equal charge density at the boundary

When $\rho_s = 0$.

$$\boxed{D_{n1} = D_{n2}}$$

**Refraction**




$$E_{t1} = E_{t2}$$

$$E_1 \sin\alpha_1 = E_2 \sin\alpha_2$$

$$D_{n1} = D_{n2}$$

$$D_1 \cos\alpha_1 = D_2 \cos\alpha_2$$

$$\varepsilon_1 E_1 \cos\alpha_1 = \varepsilon_2 E_2 \cos\alpha_2$$

$$\boxed{\frac{\tan\alpha_1}{\tan\alpha_2} = \frac{\varepsilon_1}{\varepsilon_2}}$$

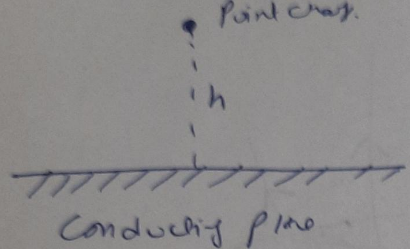
$$\boxed{\frac{E_2}{E_1} = \frac{\sin\alpha_1}{\sin\alpha_2}}$$

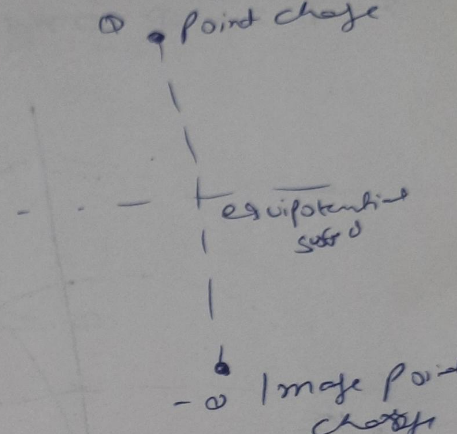
$$\vec{E} = \vec{E_t} + \vec{E_n}$$

$$\vec{D} = \vec{D_t} + \vec{D_n}$$

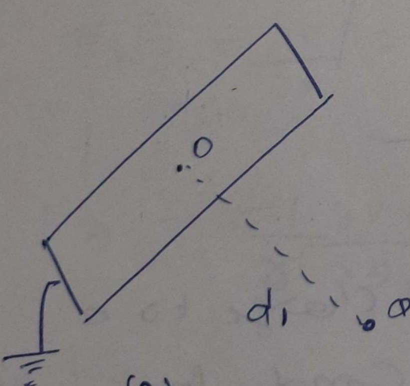
# Method of Images

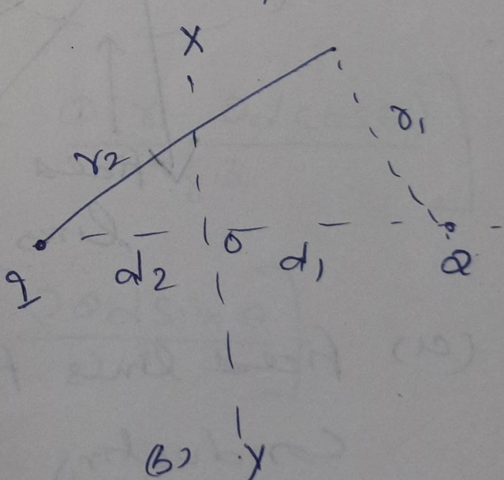
It is used for solving electrostatics field problems especially the problem Involving point charges and the Conducting plane.




(a) Images of Point charge in conducting plane



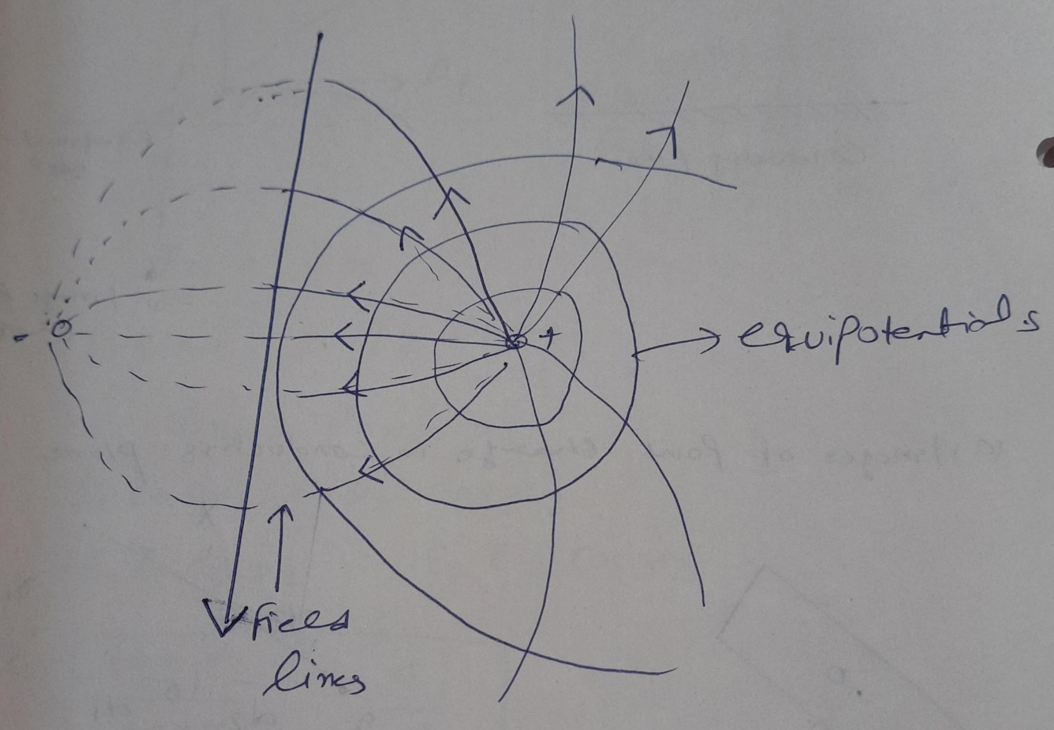




(a) Point Charge Q near the Ground Conducting plane

(b) Image Charge of Q as q

$$V = \frac{1}{4\pi\varepsilon_0}\left[\frac{Q}{r_1} + \frac{q}{r_2}\right]$$

In order to have $V = 0$, the simplest possibility is $q = -Q$ and $r_2 = r_1$ i.e $d_2 = d_1$

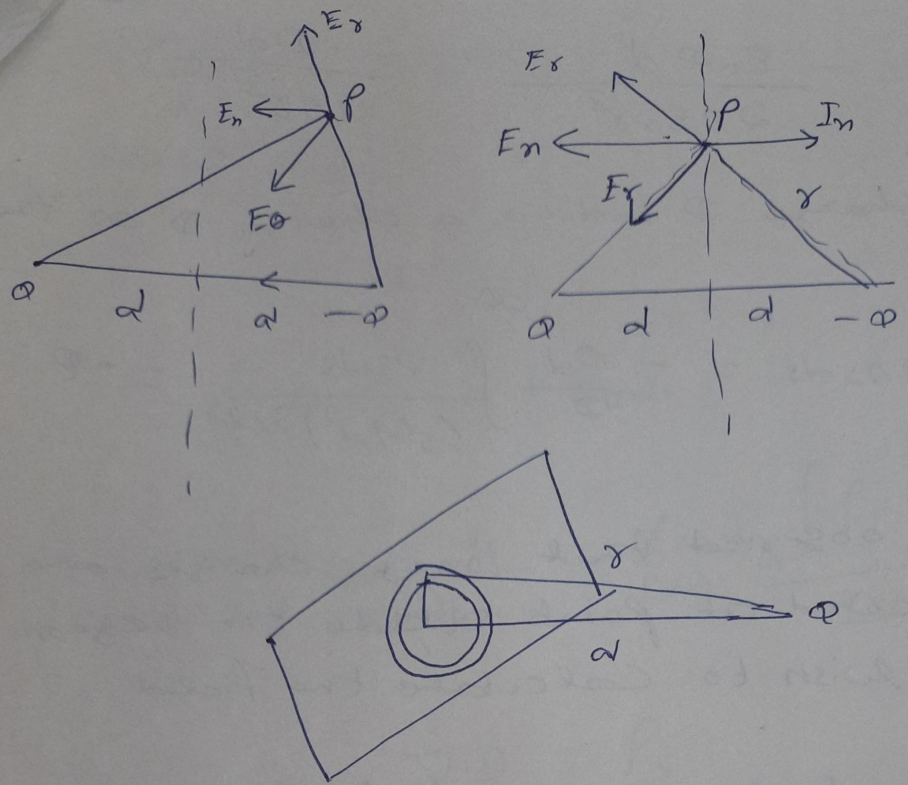



(a) field lines from point charge to a conducting grounded conductor

$$V = \frac{1}{4\pi\varepsilon_0}\left[\frac{Q}{r} - \frac{Q}{r'}\right]$$

$$r' = \sqrt{(2d - r\cos\theta)^2 + (r\sin\theta)^2}$$

$$r' = \sqrt{r^2 + 4d^2 - 4dr\cos\theta}$$




(a) Diagram for Calculating the field at a point and Charge on the Conductor

$$E_r = -\frac{\partial V}{\partial r} = \frac{1}{4\pi\varepsilon_0}\left[\frac{Q}{r^2} - \frac{Q(r - 2d\cos\theta)}{r'^3}\right]$$

$$E_\theta = \frac{-1}{r}\frac{\partial V}{\partial \theta} = \frac{-1}{4\pi\varepsilon_0}\left[\frac{2Qd\sin\theta}{r'^3}\right]$$

with $r = r'$

$$E_n = 2E_r\cos\theta = \frac{2Q\cos\theta}{4\pi\varepsilon_0 r^2} = \frac{Qd}{2\pi\varepsilon_0 r^3}$$

$$\rho_s = [-\varepsilon_0 E_n I_n] . I_n$$

$$\rho_s = -\frac{\varepsilon_0 Q d}{2\pi \varepsilon_0 r^3} = -\frac{Qd}{2\pi r^3}$$

A point charge $Q$ induces a charge $Q'$ on the conductor

$$Q' = \int_0^\infty \rho_s \, 2\pi s \, ds = -\frac{Qd}{2} \int_0^\infty \frac{2s\,ds}{(s^2+d^2)^{3/2}} = -Q$$

It will be observed that Image charges are always located at points outside the region where we wish to calculate the field.

The force between conductor and the point charge is equal to force between the point charge and its Image and is given by

$$F = \frac{1}{4\pi\varepsilon_0} \frac{Q^2}{(2d)^2}$$

## Poisson's and Laplace equations

### Gauss theorem

$$\oint D \cdot ds = Q$$

$$\oint D \cdot ds = \int \rho_v dv$$

$$\lim_{\Delta v \to 0} \frac{\int_{\Delta s} D \cdot ds}{\Delta v} = \lim_{\Delta v \to 0} \frac{\int_{\Delta v} \rho_v \cdot dv}{\Delta v}$$

So $\nabla \cdot D = \rho_s$

In a charge free region $= \boxed{\nabla \cdot D = 0}$

It is a point form of Gauss law

$$\nabla \cdot D = \rho_v$$

$$\nabla \cdot E = \frac{\rho_v}{\varepsilon_0}$$

$$E = -\nabla V$$

$$-\nabla \cdot E = \nabla \cdot \nabla V = \frac{-\rho_v}{\varepsilon_0}$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = \frac{-\rho_v}{\varepsilon}$$

It is called Poisson equation

In a charge free space $\rho_v = 0$

$$\nabla^2 V = 0$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

---

In cylindrical coordinate system

$$\nabla^2 V = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

In spherical coordinate system

$$\nabla^2 V = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial V}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial V}{\partial \theta}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial^2 V}{\partial \phi^2}$$